

Understanding Size Distributions during Lipid Nanoparticle Manufacturing through Mechanistic Modeling

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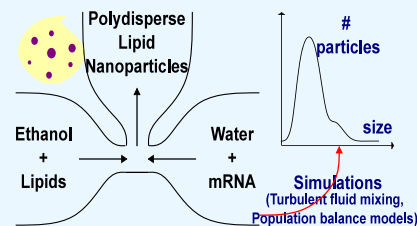


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ABSTRACT: Recent breakthroughs in messenger RNA (mRNA) therapeutics have highlighted the importance of lipid nanoparticles (LNPs) as delivery vehicles that protect fragile mRNA and facilitate cellular uptake. While manufacturing technologies that rely on self-assembly through mixing, including microfluidic systems and turbulent jet mixers, have enabled scalable production, a first-principles model explaining the fundamental mechanisms responsible for LNP size and its distribution has remained elusive. This has limited predictive process control and optimization. To address this gap, we present here a model based on crystallization of a lipid during turbulent mixing of water–ethanol, employing population balance equations to predict particle size distributions. The model successfully predicts the influence of key operating parameters including total flow rate, flow rate ratio, and lipid concentration, demonstrating favorable comparison with experimental results and enabling the development of predictive manufacturing processes for consistent critical quality attributes. The mechanistic understanding provides the foundation necessary to optimize manufacturing of LNPs used as delivery agents of mRNA vaccines, which extends beyond the current application to the broader field of nucleic acid therapeutics.



1. INTRODUCTION

Recent breakthroughs in vaccinology have underscored the therapeutic promise of mRNA (mRNA) across a variety of applications, including viral vaccines,¹ cancer treatment,² and genome editing,³ among others. The remarkable success of mRNA vaccines against COVID-19 has further fueled research into using mRNA to combat other diseases such as rabies and influenza.^{4,5} Central to this success are lipid nanoparticles (LNPs), which serve as delivery vehicles that encapsulate and shield the fragile mRNA from enzymatic breakdown while facilitating efficient cellular uptake. This ensures the release of mRNA payload into the cytoplasm where protein translation occurs.^{6,7} The physicochemical properties of LNPs, including their size, shape, and surface charge, play a crucial role in determining how effectively they cross biological barriers, enter cells, and reach their intended target organs.^{8–11} Recognizing this fundamental importance, regulatory agencies such as the World Health Organization (WHO) and the U.S. Food and Drug Administration (FDA) have designated the size and size distribution of LNPs as critical quality attributes (CQAs) for mRNA vaccine products, establishing a regulatory framework that ensures LNP-based vaccines meet consistent performance, efficiency, and safety standards.¹²

Modern LNP manufacturing predominantly relies on self-assembly through mixing. These techniques have evolved from historical techniques such as thin-film hydration, reverse-phase evaporation, high-pressure homogenization, and ultrasonication to meet large-scale production with high encapsulation

efficiency while ensuring homogeneous particle size distributions.^{6,13} During mixing, an aqueous solution containing mRNA is mixed with an ethanol solution containing lipids as shown in Figure 1a. The ethanol solution is composed of different kinds of lipids. Ionizable lipids, which are positively charged at low pH interact electrostatically with the negatively charged mRNA, thereby enhancing encapsulation efficiency of the mRNA.¹⁴ Phospholipids and cholesterol primarily enhance thermal stability and bioselectivity of the LNPs,⁶ whereas polyethylene glycol (PEG)-lipids predominantly affect size distribution.^{4,15} The most widely employed mixing approaches include microfluidic systems operating under laminar flow conditions with mixing occurring primarily through diffusion or passive mixing. However, scaling up the throughput of LNPs in microfluidic mixers has been challenging.¹⁶ On the other hand, turbulent jet mixers operate at higher Reynolds numbers, which enables rapid mixing through elevated energy dissipation, consistently producing smaller particles with narrower size distributions compared to microfluidic systems.

Despite these advances in manufacturing technology, a comprehensive mechanistic model explaining LNP self-

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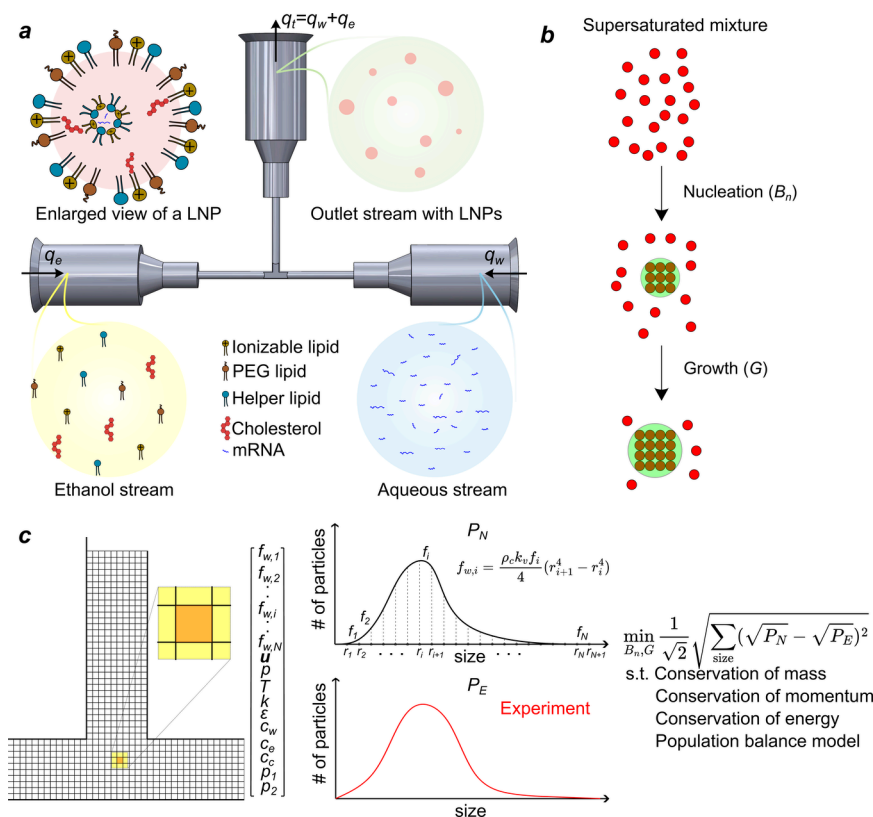


Figure 1. (a) Schematic illustration of self-assembly of different kinds of lipids (dissolved in ethanol) and mRNA (dissolved in water) to form LNPs in an impinging jet mixer. (b) Illustration of supersaturation as a driver for nucleation and growth of nanoparticles, which are the two mechanisms considered in our population balance model to predict size distribution. (c) Schematic showing our approach to estimate the rates of nucleation and growth. Using CFD simulations, the PSDs are predicted at the outlet of the impinging jet mixer. These results are compared to experimental PSDs (P_E , shown in red), with the optimal rates of nucleation and growth estimated using Bayesian Markov Chain Monte Carlo simulations by minimizing the Hellinger distance of P_N and P_E .

assembly over the range of conditions encountered in microfluidic and turbulent mixers, where the quality of water–ethanol mixing varies, has remained elusive. In particular, an aspect that is not understood very well is how the self-assembly and growth of LNPs vary in response to changes in thermal, flow and mixing conditions, which can be influenced by modifications in manufacturing processes. This has created a gap in understanding the fundamental mechanisms which explain how the characteristics of LNPs vary in response to key operating parameters such as total flow rate ($TFR = q_t$ in Figure 1a), aqueous to ethanol flow rate ratio ($FRR = q_w/q_e$ in Figure 1a) and concentration of lipids in the ethanol solution. This limitation represents a significant obstacle to predictive control of CQAs and scalable process optimization. To address this gap, here we introduce a first-principles multiscale model by drawing parallels between self-assembly of lipids into LNPs driven by a polarity shift of the solvent, and precipitation of crystals due to addition of an antisolvent during crystallization. The model includes thermal effects due to water–ethanol mixing because excess enthalpy creates temperature variations ≥ 10 K during mixing,^{17,18} which in turn affects solubility levels. We also consider turbulent mixing of the streams by leveraging advanced turbulence and micromixing models. However, the effects of mRNA-ionizable lipid and lipid–lipid interactions are neglected for the sake of simplicity. To describe spatial evolution of nanoparticle size distribution due to nucleation and growth, the population balance equations developed in¹⁹ are leveraged, as illustrated

schematically in Figure 1b. Using Bayesian Markov Chain Monte Carlo (MCMC) simulations and comparing with particle size distributions (PSDs) measured experimentally using dynamic light scattering, the optimal growth and nucleation rates of the nanoparticles are estimated as a function of lipid concentration and temperature. Finally, we use our model to understand the influence of TFR and FRR on PSDs, and obtain results consistent with experimental trends. This study opens avenues to further optimize LNP production, not just for mRNA vaccines but also for the broader fields of nucleic acid therapeutics and therapeutic proteins.

2. RESULTS

2.1. Developing a Model Which Can Predict Particle Size

Nucleation, growth, agglomeration, and breakage are the predominant mechanisms affecting PSDs in LNP self-assembly.²⁰ Among these, kinetic analysis for a well-mixed water–ethanol solution showed that nucleation and growth play a predominant role at early times, whereas agglomeration becomes the dominant mechanism at larger time scales.¹⁹ Since the PSDs computed using our model are compared to experimental data at the outlet of an impinging jet mixer with short residence times in this paper, only nucleation and growth of nanoparticles are considered in our population balance model (PBM). Assuming homogeneous nucleation, the nucleation rate is modeled based on classical nucleation theory parametrized by a tunable prefactor A_n , whereas the growth

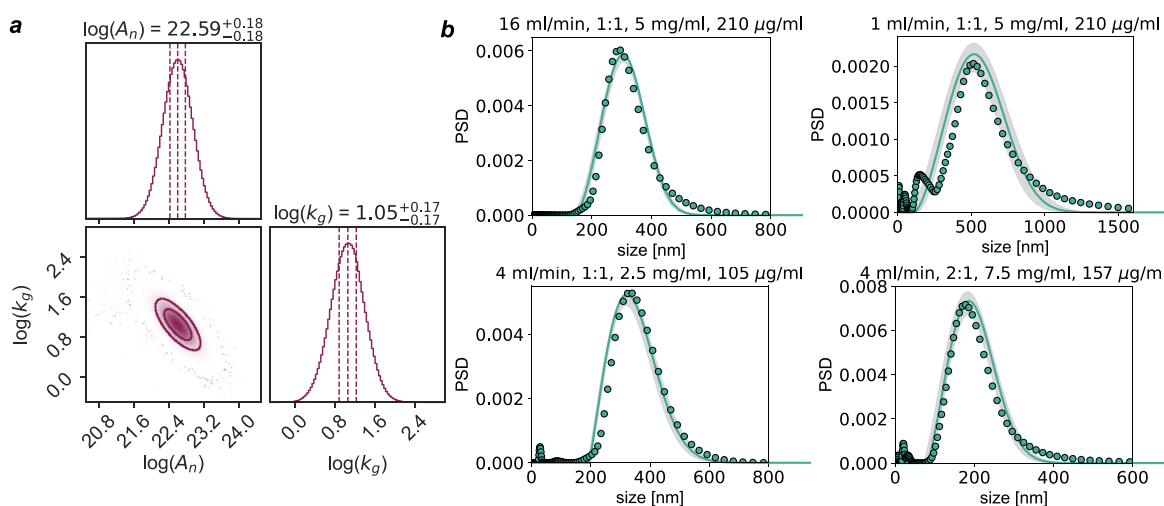


Figure 2. (a) Distributions of $\log A_n$ and $\log k_g$, where the dashed vertical lines represent the 30th, 50th, and 70th quantiles. The plot at the bottom represents the 2D histogram of the joint posterior of $\log A_n$ and $\log k_g$, where the solid lines enclose 34% and 68% confidence regions. (b) Comparison between experimental (dots) and simulated (solid line) PSDs for four representative conditions which are indicated on the top of the plot. The conditions represent the TFR, FRR, lipid concentration in ethanol, and yeast mRNA concentration in aqueous buffer, respectively. The gray region signifies a 95% confidence interval for our model.

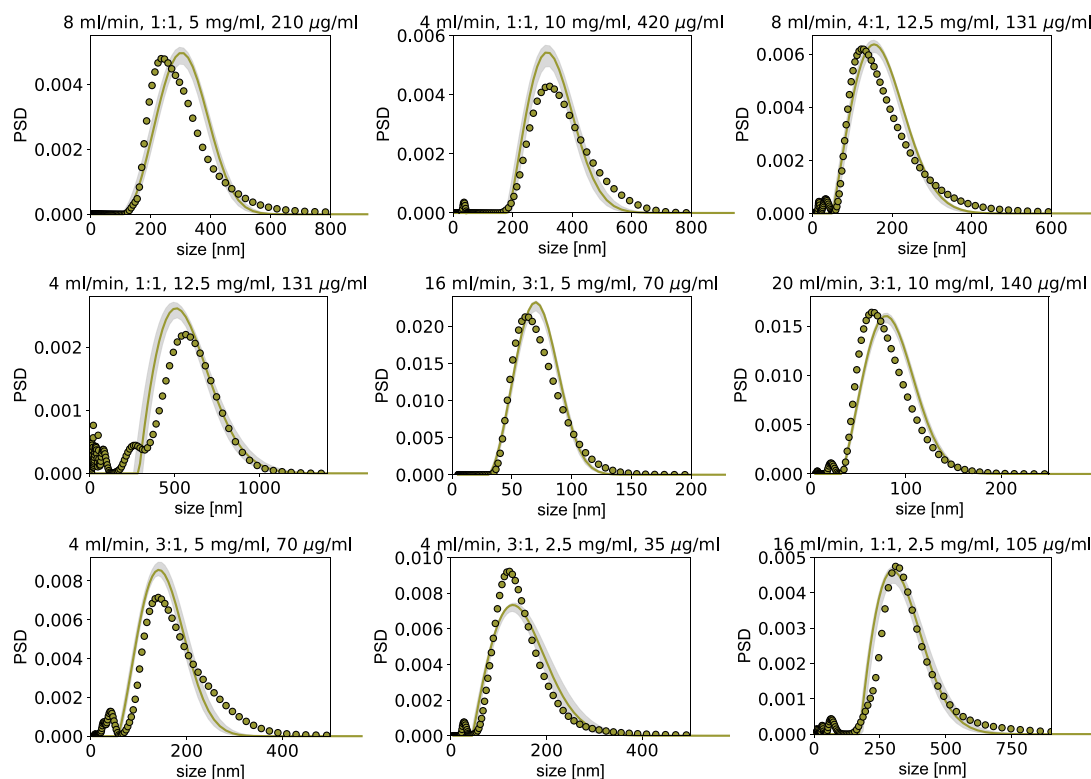


Figure 3. Comparison between experimental (dots) and simulated (solid line) PSDs for 9 representative conditions, which were not used for parameter estimation in Section 2.1. The gray region signifies a 95% confidence interval for our model.

rate is modeled using a size dependent growth model parametrized by k_g . Our objective is to estimate A_n and k_g by comparing our model predictions with experiments. The PBM is implemented together with the thermally coupled Reynolds Averaged Navier–Stokes equations and a micromixing model for water–ethanol mixing at subgrid scales; see Sec. 4.2 for a detailed explanation of our model.

To estimate the parameters A_n and k_g , spatially averaged surface integrals of the PSDs at the outlet of the impinging jet mixer are fit to experimental data using a Bayesian MCMC

method, for the conditions labeled 1–13 in Table S1 in the supplement. The likelihood is formulated as the Hellinger distance between experimental and numerically computed PSDs, as illustrated in Figure 1c, whereas uniform prior distributions $A_n \sim \mathcal{U}[10^{20}, 10^{30}]$ and $k_g \sim \mathcal{U}[0.1, 10^4]$ were taken. The adaptive Metropolis-Hastings algorithm²¹ is used to sample the posterior distributions of A_n and k_g , using 10000 samples and discard the initial burn-in of 2000 samples. Figure 2a plots the distributions of A_n and k_g and indicates the 30th,

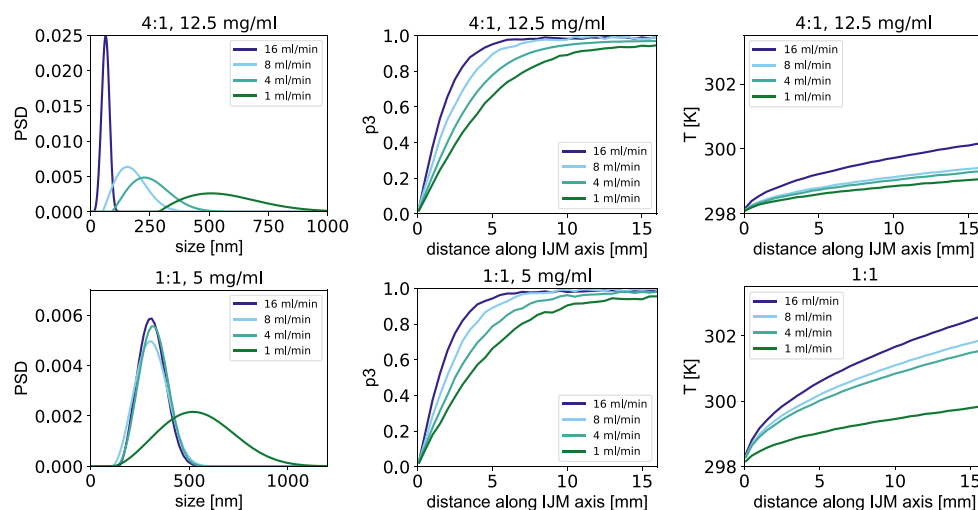


Figure 4. Effect of TFR on surface-averaged PSD, mixing efficiency and temperature along the center of the IJM, for FRRs 4:1 (top row) and 1:1 (bottom row), with lipid concentrations of 12.5 and 5 mg/mL respectively. Figure 3 shows the comparison for the intermediate FRRs 3:1 and 2:1.

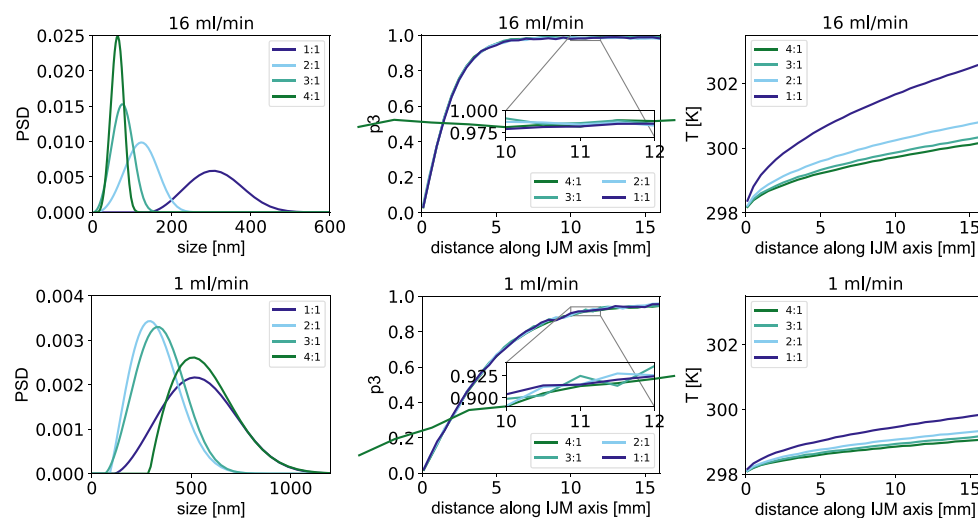


Figure 5. Effect of FRR on surface-averaged PSD, mixing efficiency and temperature along the center of the IJM, for TFR of 16 mL/min (top row) and 1 mL/min (bottom row). The lipid concentration in ethanol is 12.5, 10, 7.5, and 5 mg/mL for FRRs of 4:1, 3:1, 2:1, and 1:1, respectively. Figure 4 show the comparison for intermediate TFR of 8 and 4 mL/min.

50th, and 70th quantiles. Using the nominal values $A_n = 3.89 \times 10^{22}$ and $k_g = 11.4815$, model predictions are compared to experimental data in Figure 2b. Four representative conditions which were used to estimate the model parameters in Figure 2a and observe reasonable agreement between the experimental PSDs and 95% confidence intervals of our model predictions, with a mean $R^2 = 0.975$ for nominal A_n and k_g . For a comparison of all conditions not shown in Figure 2b, see Figure S1 in the supplement.

2.2. Model Results versus Experimental Data for New Conditions

To estimate the validity of the parameters of our model estimated in Section 2.1, model predictions and experimental data are compared for conditions not used in Section 2.1. These correspond to conditions labeled 14–26 in Table S2 in the supplement. Our model is seen to provide excellent predictions (Figure 3), with $R^2 = 0.9513$, which supports the validity of our model. In particular, good agreement occurs for a flow rate of 20 mL/min (middle row, last column in Figure 3), which is 25% greater than the maximum flow rate which

was used for training. This demonstrates the capability of the model to predict PSDs outside the range of conditions for which it was originally trained.

2.3. Effect of Operating Conditions

While estimating A_n and k_g was a primary objective of this study, a key motivation moving forward is to leverage our model to systematically understand how critical operational parameters such as TFR, FRR and lipid concentration influence PSDs, thereby providing valuable insights for nanoparticle manufacturing.

2.3.1. Effect of Total Flow Rate. Previous experimental studies have shown that increasing the TFR, while keeping the lipid concentration and FRR constant decreases the size of LNPs. Since the rate of water–ethanol mixing is enhanced at larger TFRs,^{22–24} the lipid fragments have less time to grow thereby producing smaller particles. This is observed in our model as well, as shown in Figure 4 for FRRs of 4:1 and 1:1.³⁹ While the particle size always decreases with an increase in TFR, the rate of decrease is much more pronounced at larger FRRs. To investigate this further, the surface-averaged environ-

ment probability p_3 (measure of the mixing efficiency) and surface-averaged temperature are plotted along the length of the mixer in Figure 4. The value of p_3 decreases as TFR decreases from 16 mL/min to 1 mL/min in both cases. While the values of p_3 are similar for FRRs of 4:1 and 1:1 for the same TFRs, indicating similar level of ethanol–water mixing, the temperature variation is quite different. For a TFR of 16 mL/min, a 0.74% increase in temperature is observed from the impingement point to the outlet for FRR 4:1. On the other hand, the increase is about 1.57% for 1:1. This is because in addition to conversion of turbulent kinetic energy to sensible heat, the larger heat of mixing for 1:1 compared to 4:1 further increases the temperature. This trend is consistent across all flow rates, implying a larger average temperature in the mixer as FRR decreases. The larger temperatures increase lipid solubility and therefore decrease the supersaturation of the mixture, thereby promoting nucleation and inhibiting growth of the nanoparticles. Note that, while the rate of nucleation B_n decreases with T , the decrease of supersaturation with increasing temperatures dominates in this case. Hence, we conclude that heat of mixing is the predominant reason due to which the difference in PSDs is less pronounced with varying TFR for smaller FRRs.

2.3.2. Effect of Flow Rate Ratio. Several experimental^{24–26} and numerical¹⁹ studies have shown that increasing the FRR while keeping the TFR and lipid concentration constant decreases the size of LNPs. Increasing the FRR decreases the solubility of the lipids in the water–ethanol solution, thereby leading to more nucleation and narrow PSDs. While this trend is interesting, it is not particularly informative from a manufacturing standpoint, as keeping the lipid concentration constant and increasing the FRR gives lesser yield of LNPs per unit volume of final product. Therefore, the more relevant trend in our opinion is the effect of varying the lipid concentration in ethanol at different FRR in such a way that the lipid volume fraction in the water–ethanol solution remains constant. Figure 5 shows the impact of varying FRRs for two different TFRs. For a TFR of 16 mL/min, the solution is well-mixed with $p_3 \geq 0.95$ 5 mm away from the impingement point. The variation of p_3 is relatively independent of the FRR, as shown in the insets in Figure 5. For this well-mixed case, solubility dominates the effect of heat of mixing, and particle size decreases as FRR increases. However, as the TFR decreases to 1 mL/min, a quality of mixing $p_3 \geq 0.95$ is achieved near the outlet of the mixer. Because of poor mixing, less nucleation and growth of nanoparticles occur near the impingement point and the particles continue to nucleate and grow throughout the mixer. The effect of heat of mixing also becomes more prominent, which reduces the growth of nanoparticles and promotes nucleation for smaller FRRs. Because of a combination of poor mixing and heat of mixing, the particle size is observed to not necessarily decrease as FRR increases for cases where good mixing is not achieved. Therefore, such a strategy does not seem to be a good for controlling the PSD, especially for microfluidic mixers where flow rates are small.

3. DISCUSSION AND CONCLUSIONS

A mechanistic model is proposed for predicting the particle size distribution of LNPs, which is an important quality attribute during manufacturing of mRNA-based vaccines. The model shows good agreement with in-house experiments for a range of operating conditions. The model results show that

accounting for the heat released during water ethanol mixing is important to capture the correct PSDs. Increasing the TFR generally decreases the particle size due to better mixing. However, the difference in particle size is negligible for small water–ethanol mixing ratios, where temperature rise due to the heat of mixing becomes prominent. Similarly, increasing the water–ethanol FRR decreases the particle size at large flow rates when mixing quality is good, but the trend is not consistent at small flow rates. This way, the model provides quantitative insights which can help further improve LNP manufacturing.

However, our model only predicts the formation of spherical nanoparticles and neglects lipid–lipid and lipid–mRNA interactions. Accounting for these interactions may be key to explain the formation of nanoparticles containing blebs as a function of different manufacturing conditions.^{27,28} Incorporating electrostatic interaction terms into the thermodynamic driving force for phase separation could also enable extension of the present framework to capture coupling between solvent-quality-driven aggregation and charge-mediated complexation effects, which in turn could be used to predict zeta potential and encapsulation efficiency. Although the proposed framework is general, quantitative predictions are formulation-dependent through effective solubility and kinetic parameters. Extension to other ionizable lipids (e.g., SM-102 or MC3) would require recalibration of these parameters. Nevertheless, the model provides a foundation for advanced investigation and further process optimization.

4. METHODS

4.1. Experimental Procedure to Determine PSDs

ALC-0315 ([4-hydroxybutyl]azanediyldi(hexane-6,1-diyl) bis(2-hexyldecanoate), Mw = 766.3 g/mol, 98% purity) was obtained from BroadPharm. Cholesterol (Mw = 386.65 g/mol, $\geq 99.5\%$ purity) was purchased from Sigma-Aldrich. DSPC (1,2-distearoyl-*sn*-glycero-3-phosphocholine, Mw = 790.16 g/mol, $>97.0\%$ purity) was sourced from TCI Chemicals. DMG-PEG 2000 (1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000, Mw = 2509.2 g/mol, $>99\%$ purity) was purchased from Avanti Research (formerly Avanti Polar Lipids). All lipids were used without further purification and stored at $-20\text{ }^{\circ}\text{C}$. Purified *Torula* Ambion Yeast RNA (Thermo Fisher Scientific) was used as an mRNA cargo mimic without further purification and also stored at $-20\text{ }^{\circ}\text{C}$. DNase-/RNase-free sodium acetate buffer (pH 5.5) and deionized water were obtained from Thermo Fisher Scientific. Ethanol (200 proof, molecular biology grade) was purchased from Fisher BioReagents.

Lipid stock solutions were prepared by dissolving 20 mg of total lipid mixture, comprising of ALC-0315, cholesterol, DSPC, and DMG-PEG2000 in a molar ratio of 50:38.5:10:1.5, in 100 mL of ethanol, followed by dilution to the desired concentration for each experiment. Aqueous buffer stock solutions were prepared by dissolving yeast RNA at the desired concentration in 100 mM sodium acetate buffer (pH 5.5).

Lipid nanoparticles (LNPs) were prepared by mixing the lipid solutions with aqueous yeastRNA buffer solutions in a benchtop impinging jet mixer (NanoScaler, Knauer GmbH). The jet mixer was customized with a glass window to enable optical access to the mixing chamber, allowing qualitative validation of computational flow models using tracer dyes. Prior to each experimental campaign, the system was cleaned using Triton X-100 and 2-propanol (both from Fisher Scientific). Flow rates for both inlet streams were precisely controlled, and samples were collected in glass vials. Particle size distributions were measured using a spatially resolved dynamic light scattering (SR-DLS) instrument (NanoFlowSizer, InProcess-LSP) operated in phase-sensitive detection mode (PhaSR-DLS) in batch mode. For each condition, a minimum of five full particle size distribution (PSD)

measurements were performed. All experiments were conducted at 25 °C.

4.2. Computational Model and Governing Equations

A multiscale modeling approach is used to determine the PSD of LNPs at every point in space under steady-state conditions. This modeling approach has been used before to understand size distributions of crystals formed during liquid–liquid mixing.^{29–31} Turbulent flow in the impinging jet mixer is modeled by the Navier–Stokes equations for incompressible flow with the Reynolds-Averaged Navier–Stokes (RANS) approach. The RANS equations are

$$\nabla \cdot (\rho \mathbf{u}) = 0 \quad (1)$$

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \nabla \cdot \boldsymbol{\tau}' + \rho \mathbf{g} \quad (2)$$

where $\mathbf{u}$ and p represent the Reynolds averaged velocity and pressure respectively, $\mathbf{g}$ is the acceleration due to gravity, $\boldsymbol{\tau}$ is the viscous stress, and $\boldsymbol{\tau}'$ is the Reynolds stress. The symbols ∇ and $\nabla \cdot$ represent the gradient and divergence operators in physical space. Since the mixture of water and ethanol predominantly constitute fluid in the mixer (>99% by mole fraction), Newtonian fluid behavior is assumed with no change to the viscosity of the mixture based on the viscosity of the LNPs. ρ and μ in eqs 1 and 2 are the density and viscosity of the mixture, respectively. Linear interpolation between water and ethanol properties is used for ρ and a smooth nonlinear fit for μ , which is crucial to accurately model Kelvin–Helmholtz instabilities during mixing at the water–ethanol interface as highlighted in previous simulations of water–ethanol mixing.^{17,18} Using φ to represent the mass-fraction of ethanol in the mixture, the properties and their values have been summarized in Table S1 in the Supporting Information. The Reynolds stress in eq 2 is modeled using the Boussinesq eddy viscosity approximation. The RNG k - ε turbulence model³² is used to account for subgrid scale turbulent effects, where the governing equations are

$$\nabla \cdot (\rho k \mathbf{u}) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + G_k - \rho \varepsilon \quad (3)$$

$$\nabla \cdot (\rho \varepsilon \mathbf{u}) = \nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + C_{1\varepsilon} \frac{\varepsilon}{k} G_k - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} \quad (4)$$

where μ_t is the turbulent viscosity and $C_{1\varepsilon}$, $C_{2\varepsilon}$, σ_k , σ_ε , η_0 , and β are constants; for values see Table S1 in the Supporting Information. Logarithmic law based wall functions were used to correct for the dominance of viscous effects over turbulent effects in near wall regions.³³

Water–ethanol mixing is modeled using the finite-mode probability distribution function (PDF) model proposed in ref 34, which has also been used in previous work on problems involving coupled turbulent flow and mixing.^{29,30} In this approach, the PDF describing the composition of the mixture is discretized into a finite number of Dirac delta functions in each computational cell of the CFD grid, implying

$$f_\phi(\boldsymbol{\psi}; \mathbf{x}, t) = \sum_{n=1}^{N_e} p_n(\mathbf{x}, t) \prod_{\alpha=1}^{N_s} \delta[\psi_\alpha - \langle \phi_\alpha \rangle_n(\mathbf{x}, t)] \quad (5)$$

where f_ϕ is the joint PDF of all scalars, N_e is the total number of environments, N_s is the total number of transported species, p_n represents the probability of environment n , and $\langle \phi_\alpha \rangle_n$ is the mean composition of species ϕ_α in environment n . This work uses the 3-environment model, where environment 1 represents pure water, environment 2 represents pure ethanol, and environment 3 represents the water–ethanol mixture as it aligns most naturally to the mixing process under consideration.^{35,36} The transport equations for the environment probabilities and species concentrations are

$$\rho \mathbf{u} \cdot \nabla \mathbf{p} = \nabla \cdot (\rho D_t \nabla \mathbf{p}) + \rho \mathbf{G} \quad (6)$$

$$\rho \mathbf{u} \cdot \nabla (p_3 \langle \phi \alpha \rangle_3) = \nabla \cdot [\rho D_t \nabla (p_3 \langle \phi \alpha \rangle_3)] + \rho M \quad (7)$$

where $\mathbf{p} = \{p_1, p_2\}$ is the vector of environment probabilities. $\mathbf{G}$ and M are source terms that quantify the effect of water–ethanol micromixing on transport of environment probabilities and scalar transport respectively; for full expressions see Table S1 in the Supporting Information. Turbulent diffusion of the transported species is assumed to have a diffusion coefficient $D_t = \mu_t / (\rho Sc_t)$, where Sc_t is the turbulent Schmidt number which is set to 0.7. The transported species include water (ϕ_w), ethanol (ϕ_e), lipids in dissolved form in the water–ethanol mixture (ϕ_{Ch}), and suspended solid LNPs. This work models the mixture of lipids as a 1-component lipid, whose solubility is assumed to be equal to that of cholesterol in a water–ethanol mixture. Cholesterol solubility is used as a surrogate for the effective solubility of the lipid mixture, as it represents the least soluble component and governs the onset of solvent-shift-induced phase separation in the single-component framework adopted here.

The suspended LNPs exhibit a range of sizes, which we characterize using a PBM. The PBM in¹⁹ is leveraged but we only consider the effect of nucleation and growth of LNPs as argued in Section 2.1. Charged interactions between mRNA and lipids are neglected in this model. The simplified PBM, therefore, takes the form

$$\nabla \cdot (\mathbf{u} f) + \frac{\partial}{\partial r} (G f) = \nabla \cdot (D_t \nabla f) + B_n \quad (8)$$

where f is the number density and r is the size of LNPs, B_n is the rate of nucleation of LNPs and G is the kinetic growth rate. B_n and G are functions of cholesterol solubility in the water–ethanol mixture (χ). Using experimental data from,³⁷ the solubility is modeled as a function of mixture composition and temperature; for exact form see Table S1 in the Supporting Information. The expression for B_n based on classical nucleation theory and assuming homogeneous nucleation is of the form

$$B_n = A_n \exp \left(-\frac{16\pi\sigma^3 V_m^2}{3(k_B T)^3 (\ln S)^2} \right) \delta(r - r_c) \quad (9)$$

where σ , V_m , and k_B denote interfacial energy, molecular volume, and Boltzmann constant, respectively; r_c is the critical size of the nucleated LNPs; and the prefactor A_n is a fitted parameter that controls the rate of nucleation, which is determined by comparing our simulation results with experimental particle size distributions. The growth of LNPs is modeled using a size-dependent growth model

$$G = k_g r \phi_{Ch} (S - S^*) \quad (10)$$

where $S^* = \exp((4\sigma V_m / (k_B T r)))$ is the equilibrium saturation and k_g is a fitted parameter. For further details on G and B_n , see ref 19. Since f varies spatially along r , the PDEs are discretized along the r -axis using the second-order central difference scheme in ref 38. The final set of discretized transport equations for the mass density of LNPs of different sizes in the general form of eq 7 is

$$\begin{aligned} \rho \mathbf{u} \cdot \nabla (p_3 f_{w,i}) - \nabla \cdot [\rho D_t \nabla (p_3 f_{w,i})] - \rho M \\ = \rho p_3 \frac{\rho_c k_v}{4(r_{i+1} - r_i)} (r_{i+1}^4 - r_i^4) \\ \times \begin{cases} \left[-G_{i+1} \left(f_i + \frac{r_{i+1} - r_i}{2} \frac{\partial f}{\partial r} \right) + G_i \right. \\ \left. \left(f_{i-1} + \frac{r_i - r_{i-1}}{2} \frac{\partial f}{\partial r} \right) + B_i \right], & S \geq 1 \\ \left[-G_{i+1} \left(f_{i+1} - \frac{r_{i+2} - r_{i+1}}{2} \frac{\partial f}{\partial r} \right) \right. \\ \left. + G_i \left(f_i - \frac{r_{i+1} - r_i}{2} \frac{\partial f}{\partial r} \right) \right], & S < 1 \end{cases} \quad (11) \end{aligned}$$

In addition to micromixing, enthalpy of water–ethanol mixing has been shown to significantly influence PSDs while manufacturing

medicines.³⁰ This work accounts for the nonlinear dependence of heat of water–ethanol mixing on the temperature in the mixer using

$$\nabla \cdot (\rho u E + p u) = \nabla \cdot (\kappa \nabla T) + \nabla \cdot (u \tau_{\text{eff}}) + S_h \quad (12)$$

where E and S_h represent the specific total energy and heat generation due to heat of mixing respectively, for formula see Table S1 in the Supporting Information.

Equations 1–4, 6, 7, 11, and 12 constitute our governing equations, which are discretized in spatial coordinates using a finite volume method.³¹ For details of the numerical algorithm, see Algorithm 1 in the Supporting Information.

■ ASSOCIATED CONTENT

Data Availability Statement

The code and the raw data used to produce the plots in the manuscript are publicly available at <https://github.mit.edu/mukher32/LipidNanoparticleCFDPBM>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.6c00100>.

Description of the symbols used in the manuscript, raw experimental data, and additional numerical results (PDF)

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Notes

The authors declare no competing financial interest.

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